

Design and Technology: Product Design

(A Level, AQA)

Design and Technology is an inspiring, rigorous and practical subject. This course encourages students to use creativity and imagination when applying an iterative design process to develop and modify designs, and to design and make prototypes that solve real world problems, considering their own and others' needs, wants, aspirations and values.

This Product Design course enables students to identify market needs and opportunities for new products, initiate and develop design solutions, and make and test prototypes.

Key features:

The first year will allow students to develop an understanding of many different materials classifications and the various methods for processing these materials. Students will also design, analyse and develop their practical skills through a range of investigation, design and make activities.

By the end of the first year and through the second year, candidates will be involved in a sustained design and make project (a Non-Exam Assessment or NEA), based on a design brief developed by the candidate. They will research, design, develop and manufacture a high-quality product which would be appropriate for its intended purpose.

Entry Requirements:

GCSE grade 4 or higher in Design Technology or other related subjects.

Course Content

Year 12	%	When?	Content
			Technical Principles Candidates will study the various materials categories, their properties, finishes, and applications including timbers, polymers, metals, composites and more. This unit also investigates manufacturing processes for all material types. We will look at modern and industrial scales of product design including design requirements and health and safety. We will investigate how to take a concept through to a final product including investigation into design rights, environmental impacts, designing for repair and other aspects. Designing and Making Principles Candidates will be given the opportunity to solve contextualised problems and develop advanced manufacture skills through a range of small design and make tasks. Candidates will discover various methods of communicating their designs and industrial and commercial viability. Throughout year 12 you will complete several small projects. These will not be assessed by the exam board but will be designed to generate and consolidate knowledge of materials, components and systems, as well as improve designing and practical skills and instil growing confidence in working independently in the workshop.

